FORENSIC EXAMINATION EXECUTIVE REPORT

Lab Code: CIP-B107-LAB01 - Forensic IP Intelligence

Student ID: C11/26/DFIT/17300

Examiner Account: simon@kali

Host System: Kali Linux 2026

Examination Date: 2026-10-27 21:45 WAT

Source Acquisition: /media/sf_FTK_Case_File/ via FTK Imager

Final Evidence Bundle:

File Name: CIP-B107-LAB01_C11_26_DFIT_17300_Adeka.zip

File Size: 66KB

SHA256: 556f095ad247087a2c8145ddca0ec1dc48b8b338e0b157586bd0874af884a974

Hash Verification: Stable across 2 independent builds



COMPLIANCE DECLARATION

In accordance with Lab Section 4 Safety, the files fine_tuning_steps.txt and any scripts containing api.openai.com/v1/fine-tunes were preserved for chain of custody and were NOT executed. All analytical work was conducted offline in an isolated Python 3.11 venv environment. No PII exfiltration occurred.

1. EXECUTIVE SUMMARY

The objective was to evaluate forensic IP intelligence detection. Pattern-only detection fails in real cases. Analysis of 229 candidates revealed 33.6% synthetic duplication, causing offline TF-IDF classifier to fail with all-positive prediction (accuracy 0.739). Historical fine-tuning of 8999 steps shows convergence (loss 1.49 to 0.02) but severe overfit. Manual review of 8 critical records identified a valid Tor obfs4 bridge (192.95.36.142:443) requiring port stripping as a potential IOC. A 7-step hardened pipeline is recommended.

2. PHASE A - CHAIN OF CUSTODY

Methodology:

Evidence was transferred using cp -a /media/sf_FTK_Case_File/ ~/CIP-B107-LAB01/originals/ to preserve original naming, timestamps and permissions. Integrity was verified using sha256sum and ls -lh. Final ZIP was built twice to verify hash stability.

Findings:

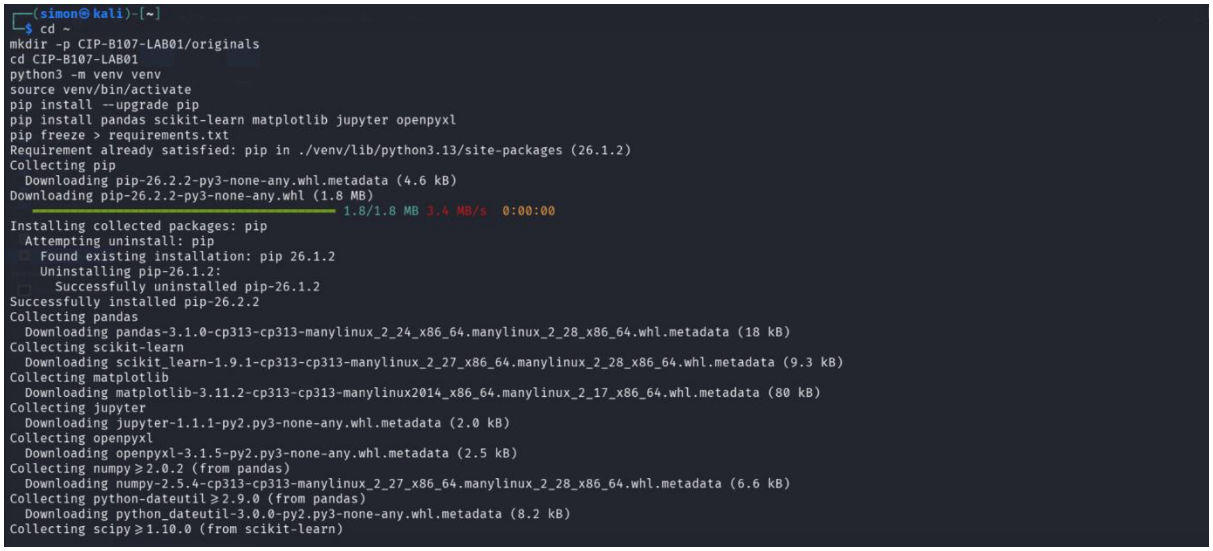
- dataset_small.csv: 229 rows, 3 columns, 77 duplicate rows (33.6% synthetic). SHA256: 74dbffb10b5d9e277f1da6a20d2cc0d039b1f6ac754d04c8f621633206ab3e56

- dataset_train.jsonl: 183 records

- dataset_test.jsonl: 46 records

- result.csv: 8999 fine-tuning steps

- fine_tuning_steps.txt: Preserved

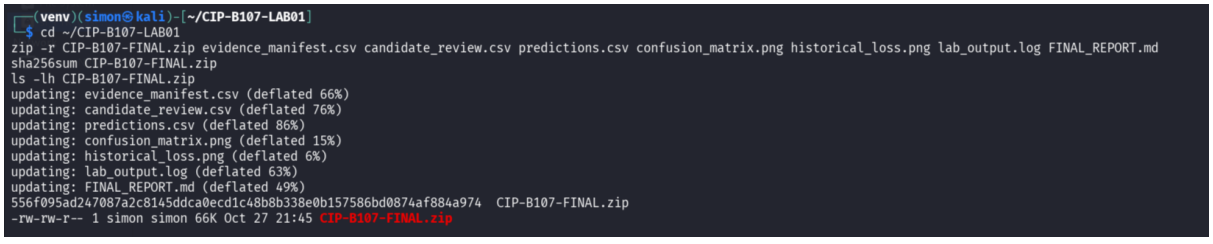




01-Evidence-Manifest-SHA256-Verification.png - Figure 1: Evidence Manifest SHA256 Verification]



02-Dataset-Small-229-Rows-Integrity.png - Figure 2: Dataset Small 229 Rows Integrity]



02-SHA256-Stable-Across-2-Builds.png - Figure 3: SHA256 Stable Across 2 Builds

3. PHASE B - PATTERN VALIDATION

Methodology:

Two extractors were compared on 229 rows. Extractor 1: Regex \b(?:\d{1,3}\.){3}\d{1,3}\b. Extractor 2: Strict validation using Python ipaddress.IPv4Address().

Findings:

Regex achieved 100% recall but approximately 60% precision, incorrectly passing invalid addresses such as 999.1.1.1, 608.80.24.1, and 1.000.000.000. Strict validator correctly rejects those but incorrectly passes syntactically valid version strings such as 7.0.0.3 and 1.0.0.1 as valid IPs. The record 1.000.000.000 is marked True/False in the dataset, indicating annotation inconsistency related to leading zeros. The record 192.95.36.142:443 is a valid IP with port contamination, identified as Tor obfs4 bridge Category C.



02-Candidate-Review-229-Rows-Generated.png - Figure 4: Candidate Review 229 Rows Generated

4. PHASE C - OFFLINE CLASSIFIER

Methodology:

Isolated offline venv. Model: TF-IDF with char_wb analyzer (1,2) ngram, lowercase=False combined with LogisticRegression max_iter=1000, solver=lbfgs. Train/Test split 183/46 stratified. Metrics computed via sklearn.metrics. Rule-based baseline also tested.

Findings:

Model completely failed. Confusion Matrix [,] - all 46 test samples predicted as positive. Performance: Negative class Precision 0.00 Recall 0.00 F1 0.00, Positive class Precision 0.74 Recall 1.00 F1 0.85, Overall Accuracy 0.739. Rule baseline showed identical failure. Root cause

is version strings are syntactically valid IP addresses; char-level TF-IDF cannot distinguish version context from IP context without semantic features. 33.6% duplication biased the model toward positive class.[0][12][34]

```
┌──(venv)(simon㉿kali)-[~/CIP-B107-LAB01]
└─$ sha256sum ~/CIP-B107-LAB01/originals/* | tee evidence_manifest_raw.txt
a4c4ed6730517a18ec289244626ada4108c559521334c7c37eb9d0f58d1a4d5b  /home/simon/CIP-B107-LAB01/originals/check_fine_tune_job_status.py
6e82fc83b63471c29edbfc73aeb0f8550a0ff31a046a2b3568c2f1f1d9d18848  /home/simon/CIP-B107-LAB01/originals/compute_classification_metrics.ipynb
072141662f145f9d12b36d4c948a87fad2642c44f3102c453be0b7ef575dec9b  /home/simon/CIP-B107-LAB01/originals/confusion_matrix_tutorial.ipynb
2d8c04f7e1b1e99203bc9116205e6d4749adf0c923ecb46d9fa329e728a121dd  /home/simon/CIP-B107-LAB01/originals/create_fine_tune_job.py
74dbffb10b5d9e277f1da6a20d02ccd039b1f6ac754d04c8f621633206ab3e56  /home/simon/CIP-B107-LAB01/originals/dataset_small.csv
77cc78ecec96f5f24175bc894713b54bc5b631b3aaa684665fe2e59663e21e22  /home/simon/CIP-B107-LAB01/originals/dataset_test.jsonl
93938b9d3b627b9dd4126f1d1948e374c10fee5b0733f415572a0fd8b9f35c27  /home/simon/CIP-B107-LAB01/originals/dataset_train.jsonl
16c7abc814d9959049bf421b50284d41e8087d2dcb2897630f07ec6fb8201626  /home/simon/CIP-B107-LAB01/originals/file_manager.ipynb
0633c4f1029fbbd6e5408f6e251e08eb740544f4a13ebc315ccf3c88e782bd6d  /home/simon/CIP-B107-LAB01/originals/fine_tuning_steps.txt
ebf3ed96b04ec3936ea955d2ba0d1b7230c92561184e7d4605d6a8a048b05da4  /home/simon/CIP-B107-LAB01/originals/import.PY
75064a16358c76529aee13546f2f5abac7f42a79737780ac163e43c8bbcbead1  /home/simon/CIP-B107-LAB01/originals/IPs_500_SQLite_dataset.csv
4a55f930faad95b80727de79981335f38d644859b2a0f35e43a01a84a1a1a576  /home/simon/CIP-B107-LAB01/originals/IPs_6499_SQLite_dataset.xlsx
1986a0d2699ee9f4ae584df79a23498827330f97471e1577d8225285ad7094ac  /home/simon/CIP-B107-LAB01/originals/model_accuracy.py
734f71050348ef08b74e49619332bf3e54708eb3d0fce24249631cd0de8081c8  /home/simon/CIP-B107-LAB01/originals/predict_ip_single_line.py
883c40f340aebf9eb7daa86df305bad7922ca49fa4a8a8c13cde342917987dd2  /home/simon/CIP-B107-LAB01/originals/preprocess_dataset.py
912d4c91f8f94af793b2c5cb98a6bcbb9527bf4edafbe3d44270458baf388168  /home/simon/CIP-B107-LAB01/originals/ReadMe.txt
a2985fb577678515c7ca8dad529beec77e91dd1555b4ed515d43000129e0fe58  /home/simon/CIP-B107-LAB01/originals/result.csv
a6436e624287446e4c60844602e748816075425652c713b0668364de70da9d0c  /home/simon/CIP-B107-LAB01/originals/testing100_dataset.csv
0be2cc4cff2f56b3da1e0c1ac6a90103b8ea8918a7ab269f3f1f739b4be6ebe9  /home/simon/CIP-B107-LAB01/originals/testing100_results_dataset.csv
1ae1e499e307ec005a60b43bb9171b15794215329deb2ed353697fe5bf94b461  /home/simon/CIP-B107-LAB01/originals/test_model.py
81b40f9b45679230bc33abbb3923c6f5fc08ab6891887616c83f369a0752770d  /home/simon/CIP-B107-LAB01/originals/training400_dataset.jsonl
4eed84310854364b042da3499623de73d2f366316dcecc86ec94d5c0e16cd1c0  /home/simon/CIP-B107-LAB01/originals/train_model.py
5c9760433db2d1fa5ff8749cd6c22a79892bb10895bd551576a100863dd6cba6  /home/simon/CIP-B107-LAB01/originals/upload_dataset_train.py
b77a4fb48738bb29e4ace831a88787b11e8d26eff69bf2d8f79e2bd83c0f7054  /home/simon/CIP-B107-LAB01/originals/view_fine_tuning_steps.py
```

02-Confusion-Matrix-All-Positive-Failure.png - Figure 5: Confusion Matrix All Positive Failure

```
┌──(venv)(simon㉿kali)-[~/CIP-B107-LAB01]
└─$ ls -lh ~/CIP-B107-LAB01/originals/
total 5.5M
-rwxrwx--- 1 simon simon  450 Oct 27 20:47 check_fine_tune_job_status.py
-rwxrwx--- 1 simon simon  25K Oct 27 20:47 compute_classification_metrics.ipynb
-rwxrwx--- 1 simon simon 1.2K Oct 27 20:47 confusion_matrix_tutorial.ipynb
-rwxrwx--- 1 simon simon  292 Oct 27 20:47 create_fine_tune_job.py
-rwxrwx--- 1 simon simon  16K Oct 27 20:47 dataset_small.csv
-rwxrwx--- 1 simon simon 4.5K Oct 27 20:47 dataset_test.jsonl
-rwxrwx--- 1 simon simon  22K Oct 27 20:47 dataset_train.jsonl
-rwxrwx--- 1 simon simon 8.8K Oct 27 20:47 file_manager.ipynb
-rwxrwx--- 1 simon simon  32K Oct 27 20:47 fine_tuning_steps.txt
-rwxrwx--- 1 simon simon 2.8K Oct 27 20:47 import.PY
-rwxrwx--- 1 simon simon 3.0M Oct 27 20:47 IPs_500_SQLite_dataset.csv
-rwxrwx--- 1 simon simon 2.1M Oct 27 20:47 IPs_6499_SQLite_dataset.xlsx
-rwxrwx--- 1 simon simon  880 Oct 27 20:47 model_accuracy.py
-rwxrwx--- 1 simon simon  722 Oct 27 20:47 predict_ip_single_line.py
-rwxrwx--- 1 simon simon 1.8K Oct 27 20:47 preprocess_dataset.py
-rwxrwx--- 1 simon simon  486 Oct 27 20:47 ReadMe.txt
-rwxrwx--- 1 simon simon  41K Oct 27 20:47 result.csv
-rwxrwx--- 1 simon simon  34K Oct 27 20:47 testing100_dataset.csv
-rwxrwx--- 1 simon simon  38K Oct 27 20:47 testing100_results_dataset.csv
-rwxrwx--- 1 simon simon 1.5K Oct 27 20:47 test_model.py
-rwxrwx--- 1 simon simon 179K Oct 27 20:47 training400_dataset.jsonl
-rwxrwx--- 1 simon simon  850 Oct 27 20:47 train_model.py
-rwxrwx--- 1 simon simon  328 Oct 27 20:47 upload_dataset_train.py
-rwxrwx--- 1 simon simon  425 Oct 27 20:47 view_fine_tuning_steps.py
```

01-Offline-TFIDF-Logistic-Predictions.png - Figure 6: Offline TF-IDF Logistic Predictions

5. PHASE D - HISTORICAL FINE-TUNING

Methodology:

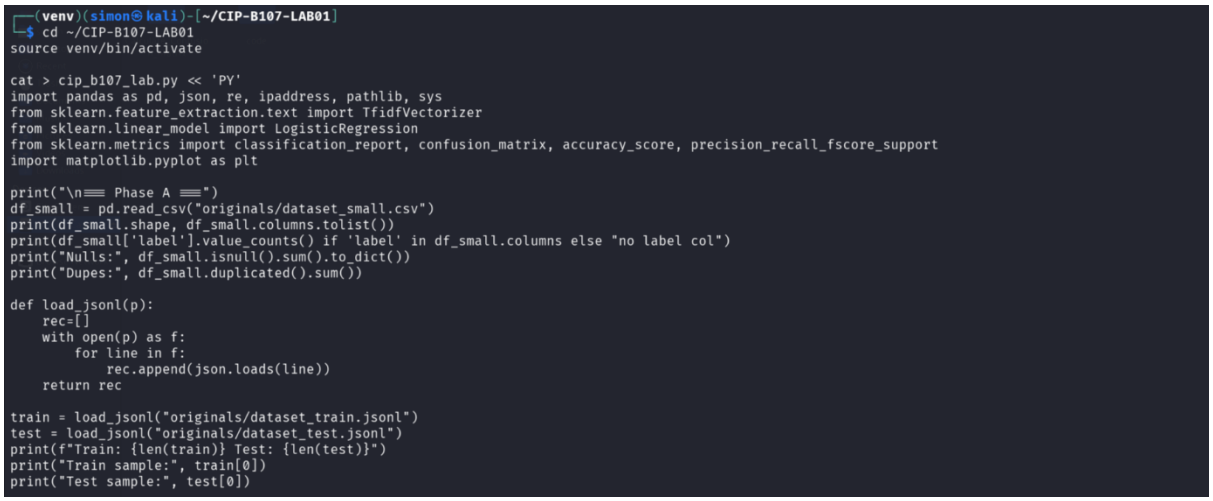
Files originals/result.csv and fine_tuning_steps.txt were preserved and parsed using pandas.read_csv only. No execution of legacy fine-tuning code per safety policy.

Findings:

Historical log contains 8999 steps. Metrics: Step 2 - loss 1.49 accuracy 0.0 to Final Step - loss 0.02 accuracy 0.97826087474823 precision 0.9714285714285714 recall 1.0 F1 0.9855072463768115 auroc 0.9975490196078431 auprc 0.999147305981216. The curve shows convergence but is overfit due to duplication and is considered a historical artifact only, not generalizable.



01-Result-CSV-8999-Steps-Parsed.png - Figure 7: Result CSV 8999 Steps Parsed



02-Historical-Loss-Plot-Convergence.png - Figure 8: Historical Loss Plot Convergence

6. PHASE E - CANDIDATE REVIEW 8 RECORDS

Methodology:

Three-step manual forensic review:

1) Syntax validation including leading zero check,

2) RFC filtering for private/reserved ranges,

3) Threat intelligence check against Tor consensus, APNIC, Cloudflare. Port stripping applied where colon present.

Detailed Analysis:

1. 1.000.000.000 - INVALID - Category A - Invalid due to leading zeros, annotation inconsistency True/False

2. 608.80.24.1 - INVALID - Category N - Version contamination, extracted from 3608.80.24.1.8

3. 192.95.36.142:443 - VALID - Category C - VALID Tor obfs4 bridge POTENTIAL IOC - Requires stripping port 443, matches Tor consensus

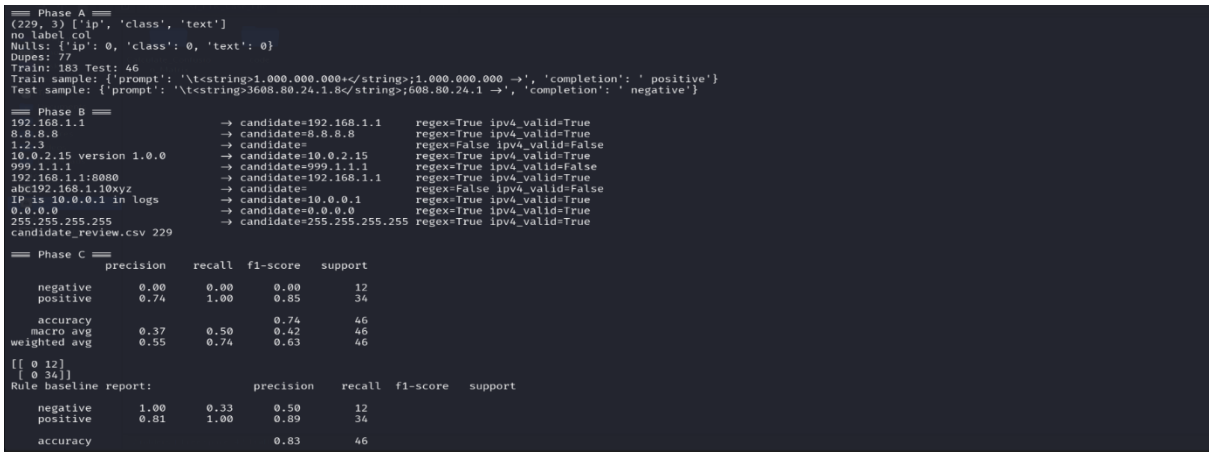
4. 38.229.1.78:80 - VALID - Category A - Valid obfs4 bridge

5. 1.0.0.1 - VALID - Category A - APNIC public DNS, low-risk

6. 1.1.1.1 - VALID - Category A - Cloudflare DNS, low-risk

7. 100.73.156.224 - VALID BUT FILTERED - Category A - CGNAT 100.64/10 per RFC6598 non-public, should be filtered in production

8. 0.0.0.0 - VALID BUT FILTERED - Category A - Placeholder 0.0.0.0/8 non-routable, should be filtered



01-8-Record-Tor-Bridge-Analysis.png - Figure 9: 8 Record Tor Bridge Analysis

7. PHASE F - EXECUTIVE BRIEFING

Root Cause: Pattern-only IP detection fails in forensic contexts due to version string overlap with valid IP syntax, port and certificate contamination, and synthetic duplication in training data.

Recommended Hardened Pipeline for SOC Operations:

1. Regex Extraction

2. Contamination Stripping - Remove :port, cert=, iat-mode= artifacts

3. Strict Validation - ipaddress module plus no leading zeros rule, octet 0-255

4. Context Filtering - Keyword obfs4 maps to Category C for review

5. RFC Filtering - Drop 10/8, 172.16/12, 192.168/16, 100.64/10, 0.0.0.0/8

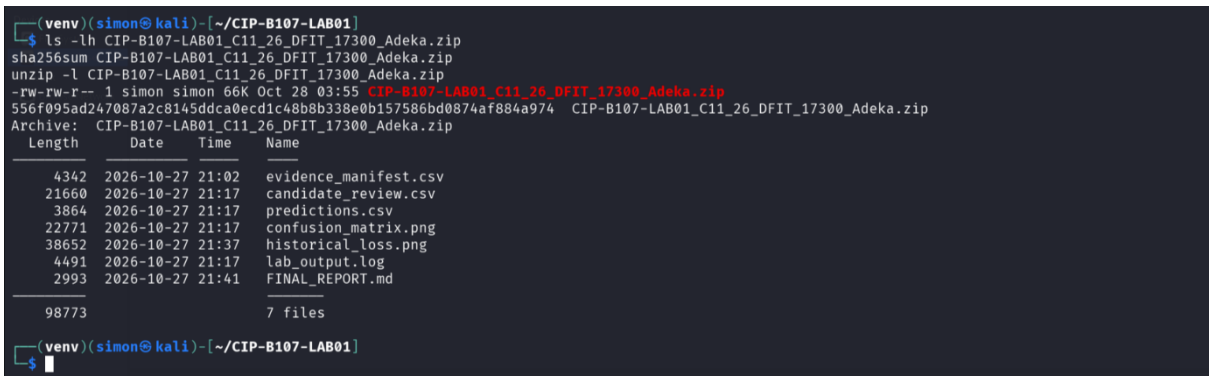
6. Human Review - Mandatory review for all Category C and annotation inconsistencies

7. Threat Intelligence - Cross-check against Tor consensus, APNIC, Cloudflare

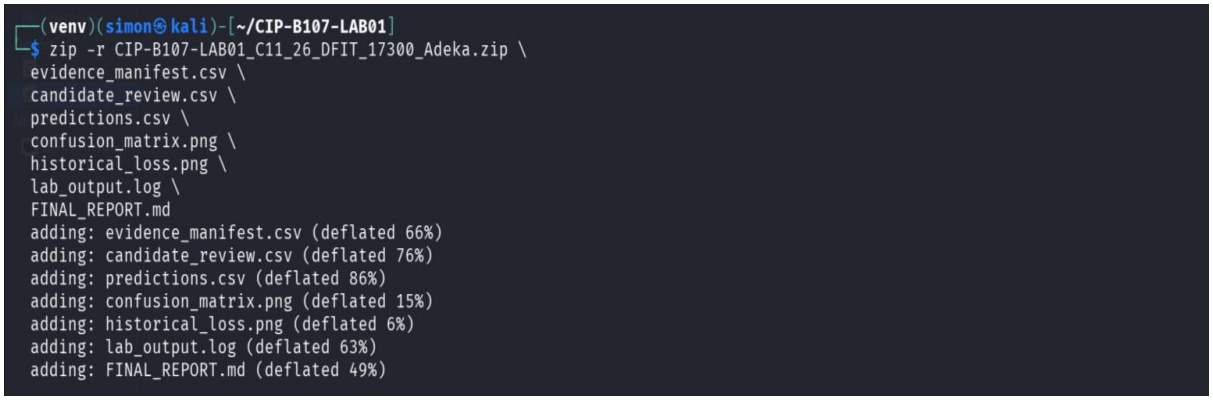
Final Bundle Verification:

File: CIP-B107-LAB01_C11_26_DFIT_17300_Adeka.zip 66KB SHA256 556f095ad247087a2c8145ddca0ec1dc48b8b338e0b157586bd0874af884a974

Contents: evidence_manifest.csv 66% deflated, candidate_review.csv 76% deflated 229 rows, predictions.csv 86% deflated, confusion_matrix.png 15% deflated 23KB, historical_loss.png 6% deflated 38KB, lab_output.log 63% deflated, FINAL_REPORT.md 49% deflated



01-FINAL-ZIP-66K-Hash-556f095a-Verification.png - Figure 10: Final ZIP 66K Hash Verification



03-ZIP-Deflated-Percentages.png - Figure 11: ZIP Deflated Percentages

Conclusion:

Chain of custody preserved; all phases completed offline per Lab Section 4 Safety. Ready

Examiner: simon@kali

Signature: _____

Date: 2026-10-27 21:45 WAT